

Linear Search

It is a simplest searching algorithm used to find an element in an array or list.

It searches the array sequentially from the beginning until the element is found or the array ends.

Working Principle

- Start from the first element
- Compare the current element with the target
- If both are equal return the index
- Otherwise move to the next element.
- Repeat until the element is found or the end of array is reached.

Algorithm

Step-1 Start from the index first

$i = 0;$

Step-2 compare current element with the index

$arr[i] == key$

Step-3 if element is found

return $i;$

Step-4 otherwise, move to next element

$i++$

Step-5 If the element is not found

return -1;

Day Run :- 2 4 8 18 20
Target = 18

Step-1

Index = 0

arr[0] = 2

$2 \neq 18$

Move to next

Step-2

Index = 1

arr[1] = 4

$4 \neq 18$

Move to next

Step-3

Index = 2

arr[2] = 8

$8 \neq 18$

move to next

Step-4

Index = 3

arr[3] = 18

$18 = 18$

Return 3

Program :-

```
for (int i = 0; i < 5; i++)
```

```
{
```

```
    if (arr[i] == key)
```

```
    {
```

```
        cout << "The element at Index : " << i;
```

```
    }
```

```
}
```

```
{
```

```
    cout << "element not found";
```

```
    return 0;
```